

IMPORTANT INSTRUCTIONS:

1. There are **TWO PARTS** in this Module/Paper. **PART ONE** contains **SHORT TYPE** questions and **PART TWO** contains **FIVE LONG TYPE** questions.
2. **PART ONE** is to be answered in the **RIGHT MARK** only, in the question paper. **PART ONE** is **NOT** to be answered in the answer book.
Single mark only. Do **NOT** double-mark on any option. Double-clicking may: Deselect your answer Or NULL answer.
3. **PART TWO** Written in answer book only provided by the invigilator.

TOTAL TIME: 3 HOURS

TOTAL MARKS: 100
(PART ONE – 40; PART TWO – 60)

PART ONE
(Answer all the questions)

Each question below gives a multiple choice of answers. Choose the most appropriate one and enter in the "OMR" answer sheet supplied with the question paper, following instructions therein. (1x35)

1. A table in the Database which belongs to 2NF but not 3NF is said to be in :
A) 1NF B) 2NF C) 3NF D) BCNF
2. Which one of the following is invalid ?
A) A table with primary key containing NULL values
B) A table with foreign key containing NULL values
C) A table with primary key containing no NULL values
D) A table with foreign key containing no NULL values
3. In respect to Boyce Codd Normal Form(BCNF) which is not true for its Functional Dependencies :
A) The left hand side contains a primary key
B) The left hand side contains a candidate key
C) The left hand side contains a super key
D) The left hand side contains a foreign key
4. Consider a schema $R(A, B, C, D, E)$ and functional dependencies $A \rightarrow B$ and $C \rightarrow D, C \rightarrow E$. - the decomposition of R into $R_1(A, B)$ and $R_2(C, D, E)$ is :
A) dependency preserving and lossless
B) lossless but not dependency preserving

- C) dependency preserving but not lossless
D) not dependency preserving and not lossless
5. In a relation $R(A, B, C, D, E)$ it is known that A, B are the candidate keys, how many super - keys are possible :
- A) 21 B) 16 C) 24 D) 23
6. Consider the following dependencies in $R(A, B, C, D, E)$, $AB \rightarrow CDE$, $B \rightarrow C$, $C \rightarrow D$, $C \rightarrow E$, $BC \rightarrow A$, $A \rightarrow B$, which can be a primary key ?
- A) A B) E C) C D) D
7. A table $R(A, B, C, D)$ is subject to partial dependency. R might be in :
- A) 1NF B) 2NF
B) 3NF D) BCNF
8. Description of data stored in database is called
- A) Schema C) Information
B) Meta Data D) None of the above
9. Which one of the following is necessarily an example of 1:1 relationship?
- A) employee-assigned_to-Department C) College-has-Principal
B) BANK-has-Branches D) None of the above
10. If $X \rightarrow Y$ holds then
- A) X determines Y uniquely C) Both A) and B) are true
B) Y determines X uniquely D) None of the above
11. The following command is used to change the structure of the table.
- A) UPDATE C) MODIFY
B) ALTER D) None of the above
12. What is the highest character taken in after a decimal in float data type ?
- A) 3 B) 4 C) 5 D) 6
13. functional Dependency is a relationship between/among
- A) Tuples B) Relations

C) Attributes

D) None of the above

14. What is the output of following code?

```
SELECT COUNT(*) FROM (SELECT NULL AS col FROM dual UNION ALL SELECT NULL FROM dual);
```

A) 0

B) 1

C) 2

D) NULL

15.

Let

$F_1 = \{ A \rightarrow B, B \rightarrow C \}$

$F_2 = \{ A \rightarrow B, A \rightarrow C \}$

Are F_1 and F_2 equivalent?

A) Yes

B) No

C) Only $F_1 \subseteq F_2$

D) Only $F_2 \subseteq F_1$

16. What is the out put of following syntax?

Create table A (id int);

Alter table A drop column id;

A) Column id drop successfully.

B) You have an error in your SQL syntax.

C) You can't delete all columns with alter table.

D) Nothing happens.

17. What is the out put of following syntax?

Create table A (id int not null auto_increment);

A) Query ok.

B) You have an error in your SQL syntax.

C) Query ok but did not show output.

D) Something else.

18. A table can have only one

1.primary key

2.auto_increment

3.not null

4. check

A) 1,2,3,4

B) 1,2

C) 1,4

D) 1

19. Which syntax did not throw error?

- A) Create table A (id int not null primary key auto_increment);
- B) Create table A (id float not null primary key auto_increment);
- C) Create table A (id varchar(20) null primary key auto_increment);
- D) Create table A (id blob not null primary key auto_increment);

20. Create table A (id int not null primary key auto_increment default 'Learning hub');

- A) Query ok.
- B) Sql syntax error.
- C) Nothing happened.
- D) Query ok but gives warning.

21. How is ER diagram represented?

- A) Circle
- B) Ellipse
- C) Triangle
- D) Square

22. Which of the following is not a SQL command?

- A) DELETE
- B) ORDER BY
- C) SELECT
- D) WHERE

23. In SQL, Wildcard characters are used with the operator.

- A) LIKE
- B) RANGE
- C) BETWEEN
- D) None of the above

24. In SQL the word 'natural' can be used with

- A) Inner join
- B) Left join
- C) Right join
- D) All of the above

25. In SQL the statement select * from R,S is equivalent to

- A) Select * from R natural join S
- B) Select * from R cross join S
- C) Select * from R union join S
- D) Select * from R inner join S

26. In SQL, which command(s) is(are) used to change a table's storage characteristics?

- A) ALTER TABLE
- B) MODIFY TABLE
- C) CHANGE TABLE
- D) All of the above

27. can be used to retrieve data from multiple table.
- A) Embedded SQL
 - B) Dynamic SQL
 - C) Joins
 - D) Views
28. NOT BETWEEN 10 AND 20
- A) displays NULL values
 - B) does not display NULL values
 - C) may display NULL values
 - D) display value
29. Which of the following is not an aggregate operator?
- A) MAX
 - B) MIN
 - C) TOTAL
 - D) AVG
30. Commit and rollback are related to
- A) data integrity
 - B) data consistency
 - C) data sharing
 - D) data security
31. Which of the following is another name for weak entity?
- A) Child
 - B) Owner
 - C) Dominant
 - D) All of the above
32. is a condition specified on a database schema and restricts the data that can be stored in an instance of the database.
- A) Key Constraint
 - B) Check Constraint
 - C) Foreign key constraint
 - D) integrity constraint
33. operations do not preserve non-matched tuples.
- A) Left outer join
 - B) Inner join
 - C) Natural join
 - D) Right outer join
34. During transaction before commit which of the following statement is done automatically in case of shutdown?
- A) Rollback
 - B) Commit
 - C) View
 - D) Flashback
35. _____ is the process of organizing data into related tables.
- A) Generalization
 - B) Normalization
 - C) Specialization
 - D) None of the above

36. There in below 30 line of code present, find which 5 lines gives error ? (1*5=5)

-- 1 DDL

CREATE DATABASE college_db;

-- 2

USE college_db;

-- 3

CREATE TABLE student (
s_id INT PRIMARY KEY,
name VARCHAR(50) NOT NULL,
age INT,
dept_id INT
);

-- 4

CREATE TABLE department (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50)
);

-- 5 DML

INSERT INTO department VALUES (1, 'CSE'), (2, 'ECE');

-- 6

INSERT INTO student VALUES (101, 'Rahul', 20, 1);

-- 7

INSERT INTO student (s_id, name, age) VALUES (102, 'Amit');

-- 8

INSERT INTO student VALUES (103, 'Priya', 21, 2);

-- 9 TCL

START TRANSACTION;

-- 10

INSERT INTO student VALUES (104, 'Neha', 22, 3);

-- 11

UPDATE student SET age = 'twenty' ;

-- 12

SAVEPOINT sp1;

-- 13

DELETE FROM student WHERE s_id = 103;

-- 14

ROLLBACK TO sp1;

-- 15
COMMIT;

-- 16 DQL
SELECT * FROM student;

-- 17
SELECT name age FROM student;

-- 18
SELECT * FROM department;

-- 19
SELECT * FORM student;

-- 20
SELECT name FROM student WHERE age > 20;

-- 21
SELECT * FROM student WHERE dept_id = 'CSE';

-- 22
UPDATE student SET age = 23 WHERE s_id = 104;

-- 23
DELETE FROM department WHERE dept_id = 2;

-- 24
SELECT * FROM department;

-- 25 TCL
START TRANSACTION;

-- 26
INSERT INTO student VALUES (105, 'Riya', 19, 1);

-- 27
ROLLBACK;

-- 28
SELECT * FROM student;

-- 29
DROP TABLE student;

-- 30
DROP DATABASE college_db;

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PART TWO
(Answer any **FOUR** questions)

37.
a) What is Normalization? Give the explanation of different normalization?
(15)
38.
a) Consider the following database tables and answer queries using SQL(not using relation algebra) Employee (emp_no, name, skill, pay_rate)
Position (posting_no, skill)
Duty-allocation (posting_no, emp_no, day, shift)
i) Get employee whose rate of pay is more than or equal to the rate of pay of employee "XYZ".
ii) Find the employee with the lowest pay rate.
iii) Get a count of different employee on each shift.
b) What is foreign key? Give a perfect example of how a foreign key connect with another schma?
(11+4)
39.
a) Draw an E-R diagram for **kalighat temple puja committee** any assumptions you make to show all the key features of the E-R modeling concept.
b) What is Join? Give the example of different type of join?
(9+6)
40.
a) Consider a relation schema R (A,B,C,D,E,G) with set F of Functional dependencies {A → B, BC → D, D → E, D → G}
Find the closure of F.
b) Consider the Universal relation R(A,B,C,D,E,F,G,H,I,J) and the set of FD's: AB → C, A → DE, B → F, F → GH, D → IJ
What is the C.key for R? Decompose R into 2NF and 3NF relations.
c) A relation R (A, B, C, D, E, F) has the following set of FD's: A → CD, B → C, F → DE, F → A
Is the decomposition of R in R1 (A, B, C), R2 (A, F, D) and R3 (E, F) dependency preserving lossless decomposition?
(5+5+5)
41. A relation R (A, B, C, D, E, F, G, H, I, J, K) has the following set of FD's:
AC → B, CE → D, EG → F, GI → H, IK → J, BD → A, DF → C, FH → E, HJ → G
What is the C.key for R?
(15)